

Unconditionally original energy-dissipative and MBP-preserving Crank-Nicolson scheme for the Allen-Cahn equation with general mobility

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ABSTRACT

In this paper, we propose a linear second-order numerical method for solving the Allen-Cahn equation with general mobility. The fully-discrete scheme is achieved by using the Crank-Nicolson formula for temporal integration and the central difference method for spatial approximation, together with two additional stabilization terms. Under mild constraints on the two stabilizing parameters, the proposed numerical scheme is shown to unconditionally preserve the discrete maximum bound principle and the discrete original energy dissipation law. Error estimate in the L^∞ norm is successfully derived for the proposed scheme. Finally, some numerical experiments are conducted to verify the theoretical results and demonstrate the performance of the proposed scheme in combination with an adaptive time-stepping strategy.

1. Introduction

The Allen-Cahn equation with general mobility is a powerful tool for modeling phase separation in materials science. By incorporating a general mobility function, more complex behaviors and dynamics that arise in real-world systems can be captured. It is well-known variable mobility can significantly affect the dynamics of phase separation, potentially leading to different interface behaviors compared to constant mobility scenarios. In this paper, we consider the following Allen-Cahn equation with a general mobility $M(\phi) \geq M_0 > 0$:

$$\begin{cases} \frac{\partial \phi}{\partial t} = -M(\phi)(-\varepsilon^2 \Delta \phi + F'(\phi)), & (\mathbf{x}, t) \in \Omega \times (0, T], \\ \phi(\mathbf{x}, 0) = \phi_0(\mathbf{x}), & \mathbf{x} \in \Omega, \end{cases} \quad (1.1)$$

where $\phi(\mathbf{x}, t)$ represents the unknown phase function, the parameter $\varepsilon > 0$ is related to the diffuse interface width, and $F(\phi) = \frac{1}{4}(1 - \phi^2)^2$ is the double-well potential function. Here, Ω is a bounded Lipschitz domain in $\mathbb{R}^d$ ($d = 1, 2, 3$) and $T > 0$ is the terminal

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time. The problem (1.1) is subject to appropriate boundary conditions, such as homogeneous Neumann, periodic, or homogeneous Dirichlet boundary conditions. The commonly used mobility in materials science applications is the so-called degenerate mobility [3,7,8,37], defined by

$$M(\phi) = (1 - \phi^2)^m, m > 0. \quad (1.2)$$

The Allen-Cahn equation (1.1) has a structure of L^2 gradient flow corresponding to the following free energy functional $E(\phi)$, defined by

$$E(\phi) = \int_{\Omega} \left(\frac{\varepsilon^2}{2} |\nabla \phi|^2 + F(\phi) \right) d\mathbf{x}, \quad (1.3)$$

which yields the following energy dissipation law

$$\frac{d}{dt} E(\phi) = - \int_{\Omega} M(\phi) \mu^2 d\mathbf{x} \leq 0, \quad (1.4)$$

where $\mu = -\varepsilon^2 \Delta \phi + F'(\phi)$. Another typical characteristic of the Allen-Cahn equation (1.1) is the maximum bound principle (MBP), which states that the solution $\phi(\mathbf{x}, t)$ satisfies $|\phi(\mathbf{x}, t)| \leq 1$ for all $\mathbf{x} \in \Omega$ and $t \geq 0$ provided the initial data $|\phi(\mathbf{x}, 0)| \leq 1$ for all $\mathbf{x} \in \Omega$. For more details, one can refer to [34]. In order to accurately simulate phase transitions and interfacial dynamics, it is essential to develop numerical methods that preserve the inherent properties of the system, such as the energy dissipation law (1.4) and the MBP. The development of structure-preserving numerical methods has played a significant role in advancing scientific discoveries and has provided valuable tools for solving complex scientific problems.

Although there has been a great deal of work [11,34,36,41,42] investigating the Allen-Cahn equation (1.1), particularly in the case of constant mobility, there is still a high demand for structure-preserving numerical methods for the general mobility model. Among the existing work, Tang et al. presented first-order linear stabilized schemes for the Allen-Cahn equation with constant mobility in [36] and the generalized problem with an advection term in [34], respectively. The unconditional MBP preservation of these proposed schemes were successfully established for both cases. However, the discrete energy dissipation of the schemes is obtained only for the case of constant mobility in [36]. In [26], a first-order linear and structure-preserving numerical method was developed for the anisotropic Allen-Cahn equation using the convex splitting approach. The proposed scheme was rigorously analyzed, establishing its unique solvability, discrete maximum principle, and unconditional energy-gradient stability. Additionally, [27] introduced first- and second-order non-iterative compact ADI-type operator splitting schemes for the Allen-Cahn equation, and the stability and accuracy of the two schemes were justified. A second order Crank-Nicolson scheme was developed for the space-fractional Allen-Cahn equation in [18], in which the discrete MBP, the energy dissipation law and the corresponding convergence analysis of the constructed numerical scheme were rigorously derived under a time step size constraint. In addition, it leads to a nonlinear system to be solved at each time step because of the use of the Crank-Nicolson scheme to deal with the nonlinear potential term. Hou and Leng [17] constructed a linear stabilized second-order Crank-Nicolson/Adams-Bashforth scheme for the Allen-Cahn equation with constant mobility. Under a constraint on the time step size, the proposed linear stabilized scheme preserves the discrete MBP and a modified energy dissipation law. In [15], a numerical study was conducted on a linear stabilized second-order backward differentiation formulation (BDF2) developed for the general Allen-Cahn equation with variable time step sizes. The discrete MBP and energy stability of the scheme were achieved through a certain constraint on the time-step size and the adjacent time ratios. Note that the energy stability means that the discrete energy can be uniformly bounded by the initial energy plus a constant. In [16], a linear doubly stabilized Crank-Nicolson scheme was investigated for the Allen-Cahn equation with general mobility, where the scheme unconditionally preserves the discrete MBP and energy stability. Moreover, the corresponding error analysis of the proposed scheme was successfully derived. However, there are still no linear second-order schemes based on the Crank-Nicolson or BDF formula that can unconditionally preserve both the discrete MBP and original energy dissipation law.

A family of MBP-preserving schemes were proposed for semilinear parabolic equations of Allen-Cahn type, based on the exponential time difference (ETD) method [4,9,20–22,25] and the integrating factor Runge-Kutta (IFRK) method [23,28,43–45]. In recent work [13], Fu et al. showed that the linear stabilized, second-order ETD Runge-Kutta (sETDRK2) scheme unconditionally preserves the dissipation of the original energy. However, these types of ETD and IFRK schemes are based on the constant variation method, which couldn't be directly generalized to problems with variable mobility. More recently, a class of multistage implicit-explicit Runge-Kutta (IMEX-RK) schemes (up to the third order in time) were designed and proven in [12] to unconditionally preserve the energy dissipation law for many gradient flow problems with Lipschitz continuous nonlinearities. It was achieved by studying the eigenvalues of matrices that depend on the IMEX-RK tableau.

The primary goal of this paper is to construct and analyze a second-order numerical scheme of Crank-Nicolson type for the Allen-Cahn equation with general mobility, which unconditionally preserves both the maximum bound principle (MBP) and the original energy dissipation law in the discrete sense. Therefore, the proposed scheme can be naturally integrated with existing time-adaptive strategies to enhance computational efficiency. The rest of paper is organized as follows. In Section 2, we present the linear doubly-stabilized Crank-Nicolson scheme on a nonuniform time grid together with the spatial approximation by the central finite difference. The unconditional preservation of the discrete MBP and original energy dissipation law are also established. In Section 3, the optimal fully-discrete error estimate in the L^∞ -norm is successfully derived. In Section 4, various numerical examples are taken to verify the theoretical results and demonstrate the performance of the proposed scheme. Some concluding remarks are given in Section 5.

2. The fully-discrete linear doubly stabilized Crank-Nicolson scheme and its properties

In this section, we first present the fully-discrete linear Crank-Nicolson scheme with two stabilized terms for the Allen-Cahn equation (1.1). Then, we investigate its unconditional preservation of the discrete MBP and the energy dissipation law. For simplicity, we will focus on the two-dimensional problem ($d = 2$) with homogeneous Neumann boundary conditions, i.e., $\frac{\partial \phi}{\partial \mathbf{n}} \Big|_{\partial \Omega} = 0$. It is important to mention that the proposed scheme and its associated analysis can be easily extended to higher-dimensional spaces and other types of boundary conditions.

2.1. The central finite difference for the Laplacian

Let's start by reviewing some basic definitions and discrete operators of the central finite difference method for the spatial discretization [1,2,19,29,35,38–40]. Denote the domain by $\Omega = (0, L_x) \times (0, L_y)$. For simplicity, we assume $L_x = L_y = L$ and the uniform spatial grid size $h = L/M$, where M represents the number of grid points in each direction. Then, two types of point sets are given by

$$\mathbf{E} = \{x_{i+\frac{1}{2}} = ih \mid i = 0, 1, \dots, M\}, \quad \mathbf{C} = \{x_i = (i - \frac{1}{2})h \mid i = 1, \dots, M\}. \quad (2.1)$$

Let $U_{i,j}$ simply denote $u(x_i, y_j)$, and the grid function spaces $C_h, e_{0,h}^x$ and $e_{0,h}^y$ be defined by

$$\begin{aligned} C_h &= \{U : \mathbf{C} \times \mathbf{C} \rightarrow \mathbb{R} \mid U_{i,j}, 1 \leq i, j \leq M\}, \\ e_{0,h}^x &= \{U \in e_h^x \mid U_{\frac{1}{2},j} = U_{M+\frac{1}{2},j} = 0, 1 \leq j \leq M\}, \\ e_{0,h}^y &= \{U \in e_h^y \mid U_{i,\frac{1}{2}} = U_{i,M+\frac{1}{2}} = 0, 1 \leq i \leq M\}. \end{aligned}$$

For any $U \in C_h$ we define the discrete gradient operator $\nabla_h = (\nabla_h^x, \nabla_h^y) : C_h \rightarrow (e_{0,h}^x, e_{0,h}^y)$ by

$$\begin{aligned} (\nabla_h^x U)_{i+\frac{1}{2},j} &= \frac{U_{i+1,j} - U_{i,j}}{h}, \quad 1 \leq i \leq M-1, 1 \leq j \leq M, \\ (\nabla_h^y U)_{i,j+\frac{1}{2}} &= \frac{U_{i,j+1} - U_{i,j}}{h}, \quad 1 \leq i \leq M, 1 \leq j \leq M-1. \end{aligned} \quad (2.2)$$

Define the grid function spaces e_h^x and e_h^y by

$$\begin{aligned} e_h^x &= \{U : \mathbf{E} \times \mathbf{C} \rightarrow \mathbb{R} \mid U_{i+\frac{1}{2},j}, 0 \leq i \leq M, 1 \leq j \leq M\}, \\ e_h^y &= \{U : \mathbf{C} \times \mathbf{E} \rightarrow \mathbb{R} \mid U_{i,j+\frac{1}{2}}, 1 \leq i \leq M, 0 \leq j \leq M\}. \end{aligned}$$

The discrete divergence operator $\nabla_h \cdot : (e_h^x, e_h^y) \rightarrow C_h$ is defined by

$$(\nabla_h \cdot (U^x, U^y)^T)_{i,j} = \frac{U_{i+\frac{1}{2},j}^x - U_{i-\frac{1}{2},j}^x}{h} + \frac{U_{i,j+\frac{1}{2}}^y - U_{i,j-\frac{1}{2}}^y}{h}, \quad 1 \leq i, j \leq M$$

for any $(U^x, U^y) \in (e_h^x, e_h^y)$. The discrete Laplacian operator $\Delta_h : C_h \rightarrow C_h$ is then correspondingly given by

$$(\Delta_h U)_{i,j} = (\nabla_h \cdot (\nabla_h U))_{i,j}, \quad 1 \leq i, j \leq M. \quad (2.3)$$

We also note that the above definition of Δ_h satisfies the discrete homogenous Neumann boundary condition. Now, we are ready to introduce some associated inner-products:

$$\begin{aligned} (U, V)_\Omega &= \langle U, V \rangle_\Omega = h^2 \sum_{i,j=1}^M U_{i,j} V_{i,j}, \quad \forall U, V \in C_h, \\ [U^x, V^x]_x &= \langle a_x(U^x V^x), 1 \rangle_\Omega, \quad \forall U^x, V^x \in e_h^x, \\ [U^y, V^y]_y &= \langle a_y(U^y V^y), 1 \rangle_\Omega, \quad \forall U^y, V^y \in e_h^y, \\ [(U^x, U^y)^T, (V^x, V^y)^T]_\Omega &= [U^x, V^x]_x + [U^y, V^y]_y, \end{aligned} \quad (2.4)$$

where the two average operators $a_x : e_h^x \rightarrow C_h$ and $a_y : e_h^y \rightarrow C_h$ are defined by

$$(a_x U)_{i,j} = \frac{1}{2}(U_{i+\frac{1}{2},j} + U_{i-\frac{1}{2},j}), \quad (a_y V)_{i,j} = \frac{1}{2}(V_{i,j+\frac{1}{2}} + V_{i,j-\frac{1}{2}}), \quad 1 \leq i, j \leq M$$

for any $U \in e_h^x, V \in e_h^y$. For any $U \in C_h$, it is straightforward to define corresponding discrete L^2 , H^1 and L^∞ norms/semi-norms as follows:

$$\begin{aligned}
\|U\|_h^2 &= \langle U, U \rangle_\Omega, \\
\|\nabla_h U\|_h^2 &= [\nabla_h U, \nabla_h U]_\Omega = [\nabla_h^x U, \nabla_h^x U]_x + [\nabla_h^y U, \nabla_h^y U]_y, \\
\|U\|_{H_h^1}^2 &= \|U\|_h^2 + \|\nabla_h U\|_h^2, \quad \|U\|_\infty = \max_{1 \leq i \leq M} \sum_{j=1}^M |U_{i,j}|.
\end{aligned} \tag{2.5}$$

There is a useful result about the discrete integration by parts.

Lemma 2.1. ([29,36]) For any $U, V \in C_h$, it holds that

$$\langle \Delta_h U, V \rangle_\Omega = -[\nabla_h U, \nabla_h V]_\Omega.$$

2.2. A linear doubly stabilized Crank-Nicolson scheme

In what follows, we consider a general partition of the time interval $[0, T]$ given by $0 = t_0 < t_1 < t_2 < \dots < t_N = T$. We denote the time step sizes by $\tau_n = t_n - t_{n-1}$ for $n = 1, 2, \dots, N$, and the maximum time step size of this time partition by $\tau = \max_{1 \leq n \leq N} \tau_n$. We define the operator Π_{C_h} as one that limits a function pointwise onto grid function space C_h . Additionally, vector $\vec{U} = [U_{1,1}, \dots, U_{1,M}; \dots; U_{M,1}, \dots, U_{M,M}]^T \in \mathbb{R}^{M^2}$ represents the vector form of $U \in C_h$.

Firstly, we recall the fully-discrete linear stabilized first-order BDF (sBDF1) scheme in [15,34] for the Allen-Cahn equation with general mobility (1.1): given $\Phi^0 = \Pi_{C_h} \phi_0$, for $n = 0, 1, \dots, N-1$, find $\Phi^{n+1} \in C_h$ such that

$$\frac{\Phi^{n+1} - \Phi^n}{\tau_{n+1}} - \varepsilon^2 M(\Phi^n) \Delta_h \Phi^{n+1} + M(\Phi^n) f(\Phi^n) + S_1 (\Phi^{n+1} - \Phi^n) = 0, \tag{2.6}$$

where $f(\Phi) = F'(\Phi)$ and $S_1 \geq 0$ is a stabilizing parameter. Let us denote the above sBDF1 scheme as $\Phi^{n+1} = \text{sBDF1}(S_1, \Phi^n, M(\Phi^n), \tau_{n+1})$. The vector form of the sBDF1 scheme (2.6) reads as

$$\frac{\vec{\Phi}^{n+1} - \vec{\Phi}^n}{\tau_{n+1}} - \varepsilon^2 \Lambda^n D_h \vec{\Phi}^{n+1} + \Lambda^n f(\vec{\Phi}^n) + S_1 (\vec{\Phi}^{n+1} - \vec{\Phi}^n) = 0, \tag{2.7}$$

where $\Lambda^n = \text{diag}(M(\vec{\Phi}^n))$ represents a diagonal matrix with the elements of vector $\vec{\Phi}^n$ as its main diagonal elements. The stiff matrix D_h is defined by $D_h = I \otimes G_h + G_h \otimes I \in \mathbb{R}^{M^2 \times M^2}$, where I denotes the identity matrix (with the matched dimensions) and G_h is a diagonally dominant tridiagonal matrix

$$G_h = \frac{1}{h^2} \begin{pmatrix} -1 & 1 & & & \\ 1 & -2 & 1 & & \\ & \ddots & \ddots & \ddots & \\ & & 1 & -2 & 1 \\ & & & 1 & -1 \end{pmatrix}_{M \times M}.$$

Now, we are ready to construct a fully-discrete linear stabilized Crank-Nicolson (sCN) scheme for the Allen-Cahn equation (1.1): given $\Phi^0 = \Pi_{C_h} \phi_0$ and for $n = 1, 2, \dots, N-1$, find $\Phi^{n+1} \in C_h$ such that

$$\begin{cases}
\vec{\Phi}^{n+\frac{1}{2}} = \text{sBDF1}(S_1, \Phi^n, M(\Phi^n), \tau_{n+1}/2), & \text{(a)} \\
\Phi^{n+\frac{1}{2}} = \text{sBDF1}(S_1 + S_2 \tau_{n+1}, \Phi^n, M(\vec{\Phi}^{n+\frac{1}{2}}), \tau_{n+1}/2), & \text{(b)} \\
\frac{\Phi^{n+1} - \Phi^n}{\tau_{n+1}} - \varepsilon^2 M(\vec{\Phi}^{n+\frac{1}{2}}) \Delta_h \frac{\Phi^{n+1} + \Phi^n}{2} + M(\vec{\Phi}^{n+\frac{1}{2}}) f(\Phi^{n+\frac{1}{2}}) \\
\quad + S_1 \left(\frac{\Phi^{n+1} + \Phi^n}{2} - \Phi^{n+\frac{1}{2}} \right) + S_2 \tau_{n+1} (\Phi^{n+1} - \Phi^n) = 0, & \text{(c)}
\end{cases} \tag{2.8}$$

where $S_2 \geq 0$ is another stabilizing parameter. The vector form of the sCN scheme (2.8) reads as

$$\begin{cases}
\vec{\Phi}^{n+\frac{1}{2}} = \text{sBDF1}(S_1, \vec{\Phi}^n, \Lambda^n, \tau_{n+1}/2), & \text{(a)} \\
\vec{\Phi}^{n+\frac{1}{2}} = \text{sBDF1}(S_1 + S_2 \tau_{n+1}, \vec{\Phi}^n, \tilde{\Lambda}^{n+\frac{1}{2}}, \tau_{n+1}/2), & \text{(b)} \\
\frac{\vec{\Phi}^{n+1} - \vec{\Phi}^n}{\tau_{n+1}} - \varepsilon^2 \tilde{\Lambda}^{n+\frac{1}{2}} D_h \frac{\vec{\Phi}^{n+1} + \vec{\Phi}^n}{2} + \tilde{\Lambda}^{n+\frac{1}{2}} f(\vec{\Phi}^{n+\frac{1}{2}}) \\
\quad + S_1 \left(\frac{\vec{\Phi}^{n+1} + \vec{\Phi}^n}{2} - \vec{\Phi}^{n+\frac{1}{2}} \right) + S_2 \tau_{n+1} (\vec{\Phi}^{n+1} - \vec{\Phi}^n) = 0, & \text{(c)}
\end{cases} \tag{2.9}$$

where $\tilde{\Lambda}^{n+\frac{1}{2}} = \text{diag}(M(\vec{\Phi}^{n+\frac{1}{2}}))$

Remark 2.1. The above sCN scheme (2.8) of second-order consists of three stages: The first stage updates $\tilde{\Phi}^{n+\frac{1}{2}}$, which is used for prediction of the variable mobility $M(\tilde{\Phi}^{n+\frac{1}{2}})$ in the subsequent two steps. The second stage updates the nonlinear term $f(\cdot)$. Finally, the third stage updates the numerical solution Φ^{n+1} . In the case of constant mobility, the first stage (2.8a) can be omitted, making the sCN scheme (2.8) equivalent to the standard two-stage unconditionally MBP-preserving CN scheme presented in [16].

2.3. Unconditional preservation of the discrete maximum bound principle

Before analyzing the discrete MBP for the CN scheme (2.8), we first present several useful lemmas.

Lemma 2.2. ([18,30,36]) Assume that $B = (b_{i,j})$ is a real $P \times P$ matrix satisfying:

$$b_{i,i} < 0, \quad |b_{i,i}| \geq \max_{1 \leq i \leq P} \sum_{j \neq i}^P |b_{i,j}|.$$

Let $A = aI - B$, where $a > 0$ is a constant, then

$$\|A\vec{U}\|_{\infty} \geq a\|\vec{U}\|_{\infty}, \quad \forall \vec{U} \in \mathbb{R}^P.$$

Lemma 2.3. If the stabilizing parameter S_1 satisfies

$$S_1 \geq \max_{\rho \in [-1,1]} \{M_*|f'(\rho)|, |(M'(\rho)f(\rho) + M(\rho)f'(\rho))|\}, \quad (2.10)$$

then it holds that

$$|S_1\rho - \alpha f(\rho)| \leq S_1, \quad \forall \alpha \in [0, M_*], \quad (2.11)$$

and

$$|S_1\rho - M(\rho)f(\rho)| \leq S_1,$$

where $M_* := \max_{\rho \in [-1,1]} |M(\rho)|$.

Proof. Let $h(\rho) = S_1\rho - \alpha f(\rho)$, then it follows from (2.10) that

$$h'(\rho) = S_1 - \alpha f'(\rho) \geq M_* \max_{\rho \in [-1,1]} |f'(\rho)| - \alpha f'(\rho) \geq 0,$$

which indicates that $h(\rho)$ is increasing with respect to $\rho \in [-1, 1]$. Together with $h(-1) = -S_1$ and $h(1) = S_1$, we have $|h(\rho)| \leq S_1$. Similarly we can obtain $|S_1\rho - M(\rho)f(\rho)| \leq S_1$, which completes the proof. $\square$

Lemma 2.4. ([34,36]) Assume that the stabilizing parameter

$$S_1 \geq \max_{\rho \in [-1,1]} |M'(\rho)f(\rho) + M(\rho)f'(\rho)|,$$

then it holds for the sBDF1 scheme (2.6) that $\|\tilde{\Phi}^{n+1}\|_{\infty} \leq 1$ for $n = 0, 1, \dots, N-1$, provided $\|\tilde{\Phi}^0\|_{\infty} \leq 1$. Particularly, if the mobility function $M(\phi) \equiv 1$, it holds that

$$E_h(\tilde{\Phi}^{n+1}) \leq E_h(\tilde{\Phi}^n)$$

for all $n = 0, 1, \dots, N-1$, provided that $S_1 \geq 2$.

The unconditional preservation of the MBP for the sCN scheme (2.9) is stated as follows.

Theorem 2.1. Assume that the stabilizing parameter S_1 satisfies (2.10) and

$$S_2 \geq \left(\frac{S_1}{4} + \frac{\varepsilon^2 M_*}{h^2} \right)^2, \quad (2.12)$$

then it holds for the sCN scheme (2.9) that $\|\tilde{\Phi}^{n+1}\|_{\infty} \leq 1$ for $n = 0, 1, \dots, N-1$, provided $\|\tilde{\Phi}^0\|_{\infty} \leq 1$.

The proof is similar to that of Theorem 2.1 in [16], thus we leave the details for interested readers.

2.4. Unconditional preservation of the discrete original energy dissipation

From the definition of the free energy $E(\phi)$ in (1.3), we define an analogous discrete energy $E_h(\Phi^n)$ as follows:

$$\begin{aligned} E_h(\Phi^n) &= \frac{\varepsilon^2}{2} [\nabla_h \Phi^n, \nabla_h \Phi^n]_\Omega + \langle F(\Phi^n), 1 \rangle_\Omega \\ &= -\frac{h^2 \varepsilon^2}{2} (\vec{\Phi}^n)^T D_h \vec{\Phi}^n + h^2 \sum_{i=1}^{M^2} F(\vec{\Phi}_i^n). \end{aligned} \quad (2.13)$$

First, we prove the discrete energy dissipation preservation of the sBDF1 scheme (2.7) for Allen-Cahn equation (1.1).

Lemma 2.5. ([31]) Let A be a $p \times p$ real symmetric matrix and satisfy:

$$a_{i,i} \geq 0, \quad |a_{i,i}| \geq \max_{1 \leq i \leq p} \sum_{j \neq i}^p |a_{i,j}|.$$

Then A is semi-positive definite.

Theorem 2.2. Assume that S_1 satisfies (2.10), then it unconditionally holds for the sBDF1 scheme (2.7) that

$$E_h(\Phi^{n+1}) \leq E_h(\Phi^n)$$

for all $n = 0, 1, \dots, N-1$.

Proof. Let $g(\vec{V}, \Lambda) = S_1(\vec{V}) - \Lambda f(\vec{V})$ for any $V \in C_h$, where Λ is a diagonal matrix and all diagonal elements belong to the interval $(0, M_*)$. Then, it is clear that

$$f(\vec{V}) = \Lambda^{-1}(S_1(\vec{V}) - g(\vec{V}, \Lambda)).$$

From the definition of the discrete energy $E_h(\cdot)$ in (2.13), we have for any $U, V \in C_h$,

$$E_h(U) - E_h(V) = h^2 \sum_{i=1}^{M^2} \left[F(\vec{U}_i) - F(\vec{V}_i) \right] - \frac{h^2 \varepsilon^2}{2} \left[(\vec{U})^T D_h \vec{U} - (\vec{V})^T D_h \vec{V} \right]. \quad (2.14)$$

Since S_1 satisfies (2.10), we have $|f'(\rho)| \leq M_*^{-1} S_1 \leq \lambda^{-1} S_1$, $\forall \rho \in [-1, 1]$, where λ is an any diagonal element of the matrix Λ . Then, for the first term in the right-hand side of (2.14), we use Taylor expansions with a Lagrange remainder to obtain that

$$\begin{aligned} h^2 \sum_{i=1}^{M^2} \left[F(\vec{U}_i) - F(\vec{V}_i) \right] &= h^2 \sum_{i=1}^{M^2} \left[f(\vec{V}_i)(\vec{U}_i - \vec{V}_i) + \frac{f'(\xi_i)}{2} (\vec{U}_i - \vec{V}_i)^2 \right] \\ &\leq h^2 \sum_{i=1}^{M^2} \left[(\vec{U}_i - \vec{V}_i) f(\vec{V}_i) + \frac{M_*^{-1} S_1}{2} (\vec{U}_i - \vec{V}_i)^2 \right] \\ &\leq h^2 (\vec{U} - \vec{V})^T f(\vec{V}) + \frac{h^2 S_1}{2} (\vec{U} - \vec{V})^T \Lambda^{-1} (\vec{U} - \vec{V}) \\ &= h^2 (\vec{U} - \vec{V})^T \Lambda^{-1} (S_1 \vec{V} - g(\vec{V}, \Lambda)) + \frac{h^2 S_1}{2} (\vec{U} - \vec{V})^T \Lambda^{-1} (\vec{U} - \vec{V}) \end{aligned} \quad (2.15)$$

For the second term of the right-hand side of (2.14), we have

$$\begin{aligned} -\frac{h^2 \varepsilon^2}{2} \left[(\vec{U})^T D_h \vec{U} - (\vec{V})^T D_h \vec{V} \right] &= -h^2 \varepsilon^2 (\vec{U} - \vec{V})^T D_h \vec{U} + \frac{h^2 \varepsilon^2}{2} (\vec{U} - \vec{V})^T D_h (\vec{U} - \vec{V}) \\ &= h^2 (\vec{U} - \vec{V})^T (S_1 \Lambda^{-1} - \varepsilon^2 D_h) \vec{U} - h^2 S_1 (\vec{U} - \vec{V})^T \Lambda^{-1} \vec{U} + \frac{h^2 \varepsilon^2}{2} (\vec{U} - \vec{V})^T D_h (\vec{U} - \vec{V}), \end{aligned} \quad (2.16)$$

where we have use the identity $a^2 - b^2 = 2(a-b)a - (a-b)^2$. Summing up (2.15) and (2.16), together with (2.14), it holds that

$$\begin{aligned} E_h(U) - E_h(V) &\leq h^2 (\vec{U} - \vec{V})^T ((S_1 \Lambda^{-1} - \varepsilon^2 D_h) \vec{U} - \Lambda^{-1} g(\vec{V}, \Lambda)) \\ &\quad - \frac{h^2}{2} (\vec{U} - \vec{V})^T (S_1 \Lambda^{-1} - \varepsilon^2 D_h) (\vec{U} - \vec{V}) \end{aligned} \quad (2.17)$$

From the definition of $g(\cdot, \cdot)$, we can use (2.7) to find

$$\left(\left(\frac{1}{\tau_{n+1}} + S_1 \right) I - \varepsilon^2 \Lambda^n D_h \right) \vec{\Phi}^{n+1} = \frac{1}{\tau_{n+1}} \vec{\Phi}^n + g(\vec{\Phi}^n, \Lambda^n).$$

Then, we have

$$g(\vec{\Phi}^n, \Lambda^n) = \left(\left(\frac{1}{\tau_{n+1}} + S_1 \right) I - \varepsilon^2 \Lambda^n D_h \right) \vec{\Phi}^{n+1} - \frac{1}{\tau_{n+1}} \vec{\Phi}^n. \quad (2.18)$$

Using (2.17), (2.18) and Lemma 2.5, we get

$$\begin{aligned} E_h(\Phi^{n+1}) - E_h(\Phi^n) &\leq h^2 (\vec{\Phi}^{n+1} - \vec{\Phi}^n)^T \left((S_1(\Lambda^n)^{-1} - \varepsilon^2 D_h) \vec{\Phi}^{n+1} - (\Lambda^n)^{-1} g(\vec{\Phi}^n, \Lambda^n) \right) \\ &\quad - \frac{h^2}{2} (\vec{\Phi}^{n+1} - \vec{\Phi}^n)^T (S_1(\Lambda^n)^{-1} - \varepsilon^2 D_h) (\vec{\Phi}^{n+1} - \vec{\Phi}^n) \\ &= h^2 (\vec{\Phi}^{n+1} - \vec{\Phi}^n)^T \left((S_1(\Lambda^n)^{-1} - \varepsilon^2 D_h) \vec{\Phi}^{n+1} \right. \\ &\quad \left. - (\Lambda^n)^{-1} \left(\left(\frac{1}{\tau_{n+1}} + S_1 \right) I - \varepsilon^2 \Lambda^n D_h \right) \vec{\Phi}^{n+1} - \frac{1}{\tau_{n+1}} \vec{\Phi}^n \right) \\ &\quad - \frac{h^2}{2} (\vec{\Phi}^{n+1} - \vec{\Phi}^n)^T (S_1(\Lambda^n)^{-1} - \varepsilon^2 D_h) (\vec{\Phi}^{n+1} - \vec{\Phi}^n) \\ &= -h^2 (\vec{\Phi}^{n+1} - \vec{\Phi}^n)^T (\Lambda^n)^{-1} \left(\left(\frac{1}{\tau_{n+1}} + \frac{1}{2} S_1 \right) I - \frac{\varepsilon^2}{2} \Lambda^n D_h \right) (\vec{\Phi}^{n+1} - \vec{\Phi}^n) \leq 0 \end{aligned}$$

which completes the proof. $\square$

Theorem 2.3. Assume that S_1 satisfies (2.10), and

$$S_2 \geq \left(\frac{S_1}{4} + \frac{2\varepsilon^2 M_*}{h^2} \right)^2, \quad (2.19)$$

then it unconditionally holds for the sCN scheme (2.9) that

$$E_h(\Phi^{n+1}) \leq E_h(\Phi^n)$$

for all $n = 0, 1, \dots, N-1$.

Proof. Similar to the proof of Theorem 2.2, we can use (2.17) and (2.9b) to deduce that

$$\begin{aligned} E_h(\Phi^{n+\frac{1}{2}}) - E_h(\Phi^n) &\leq -h^2 (\vec{\Phi}^{n+\frac{1}{2}} - \vec{\Phi}^n)^T (\tilde{\Lambda}^{n+\frac{1}{2}})^{-1} \left(\left(\frac{2}{\tau_{n+1}} + S_2 \tau_{n+1} + \frac{S_1}{2} \right) I - \frac{\varepsilon^2}{2} \tilde{\Lambda}^{n+\frac{1}{2}} D_h \right) (\vec{\Phi}^{n+\frac{1}{2}} - \vec{\Phi}^n) \end{aligned} \quad (2.20)$$

Reorganizing equation (2.9c), we can obtain

$$\begin{aligned} &\left(\left(\frac{1}{\tau_{n+1}} + S_2 \tau_{n+1} + \frac{S_1}{2} \right) I - \frac{\varepsilon^2}{2} \tilde{\Lambda}^{n+\frac{1}{2}} D_h \right) \vec{\Phi}^{n+1} \\ &= \left(\left(\frac{1}{\tau_{n+1}} + S_2 \tau_{n+1} - \frac{S_1}{2} \right) I + \frac{\varepsilon^2}{2} \tilde{\Lambda}^{n+\frac{1}{2}} D_h \right) \vec{\Phi}^n + g(\vec{\Phi}^{n+\frac{1}{2}}, \tilde{\Lambda}^{n+\frac{1}{2}}). \end{aligned} \quad (2.21)$$

From (2.18), we can derive that

$$\frac{1}{\tau_{n+1}} \vec{\Phi}^n = \left(\left(\frac{1}{\tau_{n+1}} + \frac{S_2 \tau_{n+1}}{2} + \frac{S_1}{2} \right) I - \frac{\varepsilon^2}{2} \tilde{\Lambda}^{n+\frac{1}{2}} D_h \right) \vec{\Phi}^{n+\frac{1}{2}} - \frac{S_2}{2} \tau_{n+1} \vec{\Phi}^n - \frac{1}{2} g(\vec{\Phi}^n, \tilde{\Lambda}^{n+\frac{1}{2}}) \quad (2.22)$$

Substituting (2.22) into (2.21) gives

$$\begin{aligned} g(\vec{\Phi}^{n+\frac{1}{2}}, \tilde{\Lambda}^{n+\frac{1}{2}}) &= \left(\left(\frac{1}{\tau_{n+1}} + S_2 \tau_{n+1} + \frac{S_1}{2} \right) I - \frac{\varepsilon^2}{2} \tilde{\Lambda}^{n+\frac{1}{2}} D_h \right) \vec{\Phi}^{n+1} \\ &\quad - \left(\left(\frac{1}{\tau_{n+1}} + \frac{S_2 \tau_{n+1}}{2} + \frac{S_1}{2} \right) I - \frac{\varepsilon^2}{2} \tilde{\Lambda}^{n+\frac{1}{2}} D_h \right) \vec{\Phi}^{n+\frac{1}{2}} \\ &\quad - \left(\left(\frac{1}{2} S_2 \tau_{n+1} - \frac{S_1}{2} \right) I + \frac{\varepsilon^2}{2} \tilde{\Lambda}^{n+\frac{1}{2}} D_h \right) \vec{\Phi}^n + \frac{1}{2} g(\vec{\Phi}^n, \tilde{\Lambda}^{n+\frac{1}{2}}). \end{aligned} \quad (2.23)$$

It follows from (2.17), (2.22) and (2.23) that

$$\begin{aligned}
& E_h(\Phi^{n+1}) - E_h(\Phi^{n+\frac{1}{2}}) \\
& \leq h^2(\bar{\Phi}^{n+1} - \bar{\Phi}^{n+\frac{1}{2}})^T \left((S_1(\bar{\Lambda}^{n+\frac{1}{2}})^{-1} - \varepsilon^2 D_h) \bar{\Phi}^{n+1} - (\bar{\Lambda}^{n+\frac{1}{2}})^{-1} g(\bar{\Phi}^{n+\frac{1}{2}}, \bar{\Lambda}^{n+\frac{1}{2}}) \right) \\
& \quad - \frac{h^2}{2}(\bar{\Phi}^{n+1} - \bar{\Phi}^{n+\frac{1}{2}})^T (S_1(\bar{\Lambda}^{n+\frac{1}{2}})^{-1} - \varepsilon^2 D_h) (\bar{\Phi}^{n+1} - \bar{\Phi}^{n+\frac{1}{2}}) \\
& = -h^2(\bar{\Phi}^{n+1} - \bar{\Phi}^{n+\frac{1}{2}})^T (\bar{\Lambda}^{n+\frac{1}{2}})^{-1} \left[\left(\frac{1}{\tau_{n+1}} + S_2 \tau_{n+1} \right) (\bar{\Phi}^{n+1} - \bar{\Phi}^{n+\frac{1}{2}}) + \left(\frac{1}{2} S_2 \tau_{n+1} - S_1 \right) I \right. \\
& \quad \left. + \varepsilon^2 \bar{\Lambda}^{n+\frac{1}{2}} D_h \right] \bar{\Phi}^{n+\frac{1}{2}} - \left(\left(\frac{1}{2} S_2 \tau_{n+1} - \frac{S_1}{2} \right) I + \frac{\varepsilon^2}{2} \bar{\Lambda}^{n+\frac{1}{2}} D_h \right) \bar{\Phi}^n + \frac{1}{2} g(\bar{\Phi}^n, \bar{\Lambda}^{n+\frac{1}{2}}) \Big].
\end{aligned} \tag{2.24}$$

From (2.9b), we have

$$\frac{1}{2} g(\bar{\Phi}^n, \bar{\Lambda}^{n+\frac{1}{2}}) = \left(\left(\frac{1}{\tau_{n+1}} + \frac{S_1}{2} + \frac{S_2 \tau_{n+1}}{2} \right) I - \frac{\varepsilon^2}{2} \bar{\Lambda}^{n+\frac{1}{2}} D_h \right) \bar{\Phi}^{n+\frac{1}{2}} - \left(\frac{1}{\tau_{n+1}} + \frac{S_2 \tau_{n+1}}{2} \right) \bar{\Phi}^n.$$

Substituting above equation into (2.24), we obtain that

$$\begin{aligned}
& E_h(\Phi^{n+1}) - E_h(\Phi^{n+\frac{1}{2}}) \\
& = -h^2(\bar{\Phi}^{n+1} - \bar{\Phi}^{n+\frac{1}{2}})^T (\bar{\Lambda}^{n+\frac{1}{2}})^{-1} \left(\frac{1}{\tau_{n+1}} + S_2 \tau_{n+1} \right) (\bar{\Phi}^{n+1} - \bar{\Phi}^{n+\frac{1}{2}}) \\
& \quad - h^2(\bar{\Phi}^{n+1} - \bar{\Phi}^{n+\frac{1}{2}})^T (\bar{\Lambda}^{n+\frac{1}{2}})^{-1} \left(\left(\frac{1}{\tau_{n+1}} + S_2 \tau_{n+1} - \frac{S_1}{2} \right) I + \frac{\varepsilon^2}{2} \bar{\Lambda}^{n+\frac{1}{2}} D_h \right) (\bar{\Phi}^{n+\frac{1}{2}} - \bar{\Phi}^n) \\
& = -h^2(\bar{\Phi}^{n+1} - \bar{\Phi}^{n+\frac{1}{2}})^T (\bar{\Lambda}^{n+\frac{1}{2}})^{-1} \left(\frac{1}{\tau_{n+1}} + S_2 \tau_{n+1} \right) (\bar{\Phi}^{n+1} - \bar{\Phi}^{n+\frac{1}{2}}) \\
& \quad + \frac{h^2}{2}(\bar{\Phi}^{n+1} - \bar{\Phi}^{n+\frac{1}{2}})^T (\bar{\Lambda}^{n+\frac{1}{2}})^{-1} \left(\left(\frac{1}{\tau_{n+1}} + S_2 \tau_{n+1} - \frac{S_1}{2} \right) I + \frac{\varepsilon^2}{2} \bar{\Lambda}^{n+\frac{1}{2}} D_h \right) (\bar{\Phi}^{n+1} - \bar{\Phi}^{n+\frac{1}{2}}) \\
& \quad + \frac{h^2}{2}(\bar{\Phi}^{n+\frac{1}{2}} - \bar{\Phi}^n)^T (\bar{\Lambda}^{n+\frac{1}{2}})^{-1} \left(\left(\frac{1}{\tau_{n+1}} + S_2 \tau_{n+1} - \frac{S_1}{2} \right) I + \frac{\varepsilon^2}{2} \bar{\Lambda}^{n+\frac{1}{2}} D_h \right) (\bar{\Phi}^{n+\frac{1}{2}} - \bar{\Phi}^n) \\
& \quad - \frac{h^2}{2}(\bar{\Phi}^{n+1} - \bar{\Phi}^n)^T (\bar{\Lambda}^{n+\frac{1}{2}})^{-1} \left(\left(\frac{1}{\tau_{n+1}} + S_2 \tau_{n+1} - \frac{S_1}{2} \right) I + \frac{\varepsilon^2}{2} \bar{\Lambda}^{n+\frac{1}{2}} D_h \right) (\bar{\Phi}^{n+1} - \bar{\Phi}^n),
\end{aligned} \tag{2.25}$$

where the identity $(a-b)(b-c) = -\frac{1}{2}(a-b)^2 - \frac{1}{2}(b-c)^2 + \frac{1}{2}(a-c)^2$ is used. Summing the inequalities (2.20) and (2.25) yields

$$\begin{aligned}
E_h(\Phi^{n+1}) - E_h(\Phi^n) & \leq -\frac{h^2}{2}(\bar{\Phi}^{n+1} - \bar{\Phi}^{n+\frac{1}{2}})^T (\bar{\Lambda}^{n+\frac{1}{2}})^{-1} \Delta_1 (\bar{\Phi}^{n+1} - \bar{\Phi}^{n+\frac{1}{2}}) \\
& \quad - \frac{h^2}{2}(\bar{\Phi}^{n+\frac{1}{2}} - \bar{\Phi}^n)^T (\bar{\Lambda}^{n+\frac{1}{2}})^{-1} \Delta_2 (\bar{\Phi}^{n+\frac{1}{2}} - \bar{\Phi}^n) \\
& \quad - \frac{h^2}{2}(\bar{\Phi}^{n+1} - \bar{\Phi}^n)^T (\bar{\Lambda}^{n+\frac{1}{2}})^{-1} \Delta_3 (\bar{\Phi}^{n+1} - \bar{\Phi}^n),
\end{aligned}$$

where the matrices $\Delta_i, i = 1, 2, 3$ are defined as follows:

$$\begin{aligned}
\Delta_1 & = \left(\frac{1}{\tau_{n+1}} + S_2 \tau_{n+1} + \frac{1}{2} S_1 \right) I - \frac{1}{2} \varepsilon^2 \bar{\Lambda}^{n+\frac{1}{2}} D_h, \\
\Delta_2 & = \left(\frac{3}{\tau_{n+1}} + S_2 \tau_{n+1} + \frac{3S_1}{2} \right) I - \frac{3}{2} \varepsilon^2 \bar{\Lambda}^{n+\frac{1}{2}} D_h, \\
\Delta_3 & = \left(\frac{1}{\tau_{n+1}} + S_2 \tau_{n+1} - \frac{S_1}{2} \right) I + \frac{\varepsilon^2}{2} \bar{\Lambda}^{n+\frac{1}{2}} D_h.
\end{aligned}$$

Consequently, it follows from the semi-positive definiteness of $-D_h$ that Δ_1 and Δ_2 are positive definite. Using the definition of the matrix of D_h and the condition (2.19), we can deduce that Δ_3 is a symmetric diagonally dominant matrix with non-negative diagonal elements. Together with Lemma 2.5, we can derive that Δ_3 is also semi-positive definite. Therefore, we obtain the desired result:

$$E_h(\Phi^{n+1}) - E_h(\Phi^n) \leq 0,$$

which ends the proof. $\square$

3. The L^∞ error estimate

In this section, we investigate the L^∞ -norm error estimate of the proposed sCN scheme (2.9) for the Allen-Cahn equation with general mobility. Let us define the error functions as follows:

$$e^n = \Phi^n - \Phi(t_n), \quad e^{n+\frac{1}{2}} = \Phi^{n+\frac{1}{2}} - \Phi(t_{n+\frac{1}{2}}), \quad \bar{e}^{n+\frac{1}{2}} = \bar{\Phi}^{n+\frac{1}{2}} - \Phi(t_{n+\frac{1}{2}})$$

with $\Phi(t) = \Pi_{C_h} \phi(t)$. Let C , C^* and C_i denote some generic positive constants independent of h and $\{\tau_n\}$.

Lemma 3.1 (Discrete Gronwall lemma [33]). Assume that k_n is a non-negative sequence, and the sequence ϕ_n satisfies

$$\begin{cases} \phi_0 \leq g_0 \\ \phi_n \leq g_0 + \sum_{s=0}^{n-1} k_s \phi_s, n \geq 1 \end{cases}$$

Then ϕ_n satisfies moreover, if $g_0 \geq 0$ and $p_n \geq 0$ for $n \geq 0$, it follows

$$\phi_n \leq (g_0) \exp\left(\sum_{s=0}^{n-1} k_s\right), n \geq 1.$$

Theorem 3.1. Assume S_1 satisfies (2.10), S_2 satisfies (2.12) and

$$\phi \in W^{3,\infty}(0, T; L^\infty(\Omega)) \cap L^\infty(0, T; W^{4,\infty}(\Omega)).$$

Then it holds for the sCN scheme (2.9) that

$$\|e^{n+1}\|_\infty \leq C \exp(C^* T) (\tau^2 + h^2) \quad (3.1)$$

for all $0 \leq n \leq N-1$.

Proof. According to (1.1), the exact solution $\bar{\Phi}(\cdot)$ satisfies

$$\frac{\bar{\Phi}(t_{n+\frac{1}{2}}) - \bar{\Phi}(t_n)}{\tau_{n+1}/2} + \Lambda(\bar{\Phi}(t_{n+\frac{1}{2}}))(-\varepsilon^2 D_h \bar{\Phi}(t_{n+\frac{1}{2}}) + f(\bar{\Phi}(t_{n+\frac{1}{2}}))) + \bar{T}_1^n + \Lambda(\bar{\Phi}(t_{n+\frac{1}{2}}))\bar{T}_2^n = 0, \quad (3.2)$$

where the local truncation errors $\bar{T}_1^n$ and $\bar{T}_2^n$ are defined by

$$\bar{T}_1^n = \bar{\Phi}_t(t_{n+\frac{1}{2}}) - \frac{\bar{\Phi}(t_{n+\frac{1}{2}}) - \bar{\Phi}(t_n)}{\tau_{n+1}/2}, \quad \bar{T}_2^n = -\varepsilon^2 \Delta \bar{\Phi}(t_{n+\frac{1}{2}}) + \varepsilon^2 D_h \bar{\Phi}(t_{n+\frac{1}{2}}).$$

Furthermore, it holds for $\bar{T}_1^n$ and $\bar{T}_2^n$ that

$$\|\bar{T}_1^n\|_\infty \leq \frac{\tau_{n+1}}{4} \|\phi\|_{W^{2,\infty}(0,T;L^\infty(\Omega))}, \quad \|\bar{T}_2^n\|_\infty \leq \frac{\varepsilon^2}{6} h^2 \|\phi\|_{L^\infty(0,T;W^{4,\infty}(\Omega))}. \quad (3.3)$$

We use (3.2) and (2.9a) to get the following error equation

$$\begin{aligned} \frac{\bar{e}^{n+\frac{1}{2}} - \bar{e}^n}{\tau_{n+1}/2} - \varepsilon^2 \Lambda^n D_h \bar{e}^{n+\frac{1}{2}} + S_1(\bar{e}^{n+\frac{1}{2}} - \bar{e}^n) + S_1(\bar{\Phi}(t_{n+\frac{1}{2}}) \\ - \bar{\Phi}(t_n)) - [\Lambda^n - \Lambda(\bar{\Phi}(t_{n+\frac{1}{2}}))][\varepsilon^2 D_h \bar{\Phi}(t_{n+\frac{1}{2}}) - f(\bar{\Phi}(t_{n+\frac{1}{2}}))] \\ + \Lambda^n[f(\bar{\Phi}^n) - f(\bar{\Phi}(t_{n+\frac{1}{2}}))] - \bar{T}_1^n - \Lambda(\bar{\Phi}(t_{n+\frac{1}{2}}))\bar{T}_2^n = 0, \end{aligned} \quad (3.4)$$

which leads to

$$\begin{aligned} \left(\left(1 + \frac{S_1 \tau_{n+1}}{2}\right) I - \frac{\varepsilon^2 \tau_{n+1}}{2} \Lambda^n D_h \right) \bar{e}^{n+\frac{1}{2}} \\ = \bar{e}^n + \frac{\tau_{n+1}}{2} \left[S_1 \bar{e}^n - S_1(\bar{\Phi}(t_{n+\frac{1}{2}}) - \bar{\Phi}(t_n)) - \Lambda^n(f(\bar{\Phi}^n) - f(\bar{\Phi}(t_{n+\frac{1}{2}}))) \right. \\ \left. + (\Lambda^n - \Lambda(\bar{\Phi}(t_{n+\frac{1}{2}})))(\varepsilon^2 D_h \bar{\Phi}(t_{n+\frac{1}{2}}) - f(\bar{\Phi}(t_{n+\frac{1}{2}}))) + \bar{T}_1^n + \Lambda(\bar{\Phi}(t_{n+\frac{1}{2}}))\bar{T}_2^n \right]. \end{aligned} \quad (3.5)$$

Using Lemma 2.2 and the triangle inequality, we can obtain from the above error equation that

$$\begin{aligned}
\|\tilde{e}^{n+\frac{1}{2}}\|_{\infty} &\leq \left\| \left(\left(1 + \frac{S_1 \tau_{n+1}}{2}\right) I - \frac{\varepsilon^2 \tau_{n+1}}{2} \Lambda^n D_h \right) \tilde{e}^{n+\frac{1}{2}} \right\|_{\infty} \\
&\leq \left(1 + \frac{S_1 \tau_{n+1}}{2}\right) \|\tilde{e}^n\|_{\infty} + \frac{\tau_{n+1}}{2} \left[\frac{S_1}{2} \|\phi\|_{W^{1,\infty}(0,T;L^{\infty}(\Omega))} \tau_{n+1} \right. \\
&\quad + M_* \|f(\tilde{\Phi}^n) - f(\tilde{\Phi}(t_{n+\frac{1}{2}}))\|_{\infty} + \|\Lambda^n - \Lambda(\tilde{\Phi}(t_{n+\frac{1}{2}}))\|_{\infty} \|\varepsilon^2 D_h \tilde{\Phi}(t_{n+\frac{1}{2}}) \\
&\quad \left. - f(\tilde{\Phi}(t_{n+\frac{1}{2}}))\|_{\infty} + \|\tilde{T}_1^n\|_{\infty} + M_* \|\tilde{T}_2^n\|_{\infty} \right].
\end{aligned} \tag{3.6}$$

From the definition of $F(\rho) = \frac{1}{4}(1 - \rho^2)^2$ with $f(\rho) = F'(\rho)$, it holds that

$$\max_{\rho \in [-1,1]} f(\rho) = \frac{2}{3\sqrt{3}} \quad \text{and} \quad \max_{\rho \in [-1,1]} f'(\rho) = 2.$$

Furthermore, we apply Lagrange's Mean Value Theorem to obtain

$$\begin{aligned}
\|f(\tilde{\Phi}^n) - f(\tilde{\Phi}(t_{n+\frac{1}{2}}))\|_{\infty} &\leq \|f(\tilde{\Phi}^n) - f(\tilde{\Phi}(t_n))\|_{\infty} + \|f(\tilde{\Phi}(t_n)) - f(\tilde{\Phi}(t_{n+\frac{1}{2}}))\|_{\infty} \\
&\leq 2\|\tilde{e}^n\|_{\infty} + \tau_{n+1} \|\phi\|_{W^{1,\infty}(0,T;L^{\infty}(\Omega))},
\end{aligned} \tag{3.7}$$

and

$$\begin{aligned}
\|\Lambda^n - \Lambda(\tilde{\Phi}(t_{n+\frac{1}{2}}))\|_{\infty} &\leq \|\Lambda^n - \Lambda(\tilde{\Phi}(t_n))\|_{\infty} + \|\Lambda(\tilde{\Phi}(t_n)) - \Lambda(\tilde{\Phi}(t_{n+\frac{1}{2}}))\|_{\infty} \\
&\leq \max_{\rho \in [-1,1]} |M'(\rho)| [\|\tilde{e}^n\|_{\infty} + \frac{\tau_{n+1}}{2} \|\phi\|_{W^{1,\infty}(0,T;L^{\infty}(\Omega))}].
\end{aligned} \tag{3.8}$$

From the definition of T_2^n and the triangle inequality, we have

$$\begin{aligned}
\|-\varepsilon^2 D_h \tilde{\Phi}(t_{n+\frac{1}{2}}) + f(\tilde{\Phi}(t_{n+\frac{1}{2}}))\|_{\infty} &= \|-\varepsilon^2 \Delta \tilde{\Phi}(t_{n+\frac{1}{2}}) - \tilde{T}_2^n + f(\tilde{\Phi}(t_{n+\frac{1}{2}}))\|_{\infty} \\
&\leq \varepsilon^2 \|\phi\|_{L^{\infty}(0,T;W^{2,\infty}(\Omega))} + \|\tilde{T}_2^n\|_{\infty} + \frac{2}{3\sqrt{3}} := C_1.
\end{aligned} \tag{3.9}$$

Substituting the estimates (3.7)-(3.9) into (3.6), together with (3.3), we have

$$\|\tilde{e}^{n+\frac{1}{2}}\|_{\infty} \leq \|\tilde{e}^n\|_{\infty} + C_3 \tau_{n+1} \|\tilde{e}^n\|_{\infty} + C_4 \tau_{n+1}^2 + \frac{\varepsilon^2 h^2 M_*}{12} \|\phi\|_{L^{\infty}(0,T;W^{4,\infty}(\Omega))} \tau_{n+1}, \tag{3.10}$$

where

$$\begin{aligned}
C_2 &= \frac{C_1 \max_{\rho \in [-1,1]} |M'(\rho)|}{2}, \quad C_3 = \frac{S_1}{2} + C_2 + M_*, \\
C_4 &= \frac{S_1 + 2M_* + 2C_2}{4} \|\phi\|_{W^{1,\infty}(0,T;L^{\infty}(\Omega))} + \frac{1}{8} \|\phi\|_{W^{2,\infty}(0,T;L^{\infty}(\Omega))}.
\end{aligned}$$

Similarly, we get the following error equation of $\tilde{e}^{n+\frac{1}{2}}$ from (3.2) and (2.9b)

$$\begin{aligned}
\left(\left(\frac{2}{\tau_{n+1}} + \frac{S_2 \tau_{n+1}}{2} + S_1 \right) I - \varepsilon^2 \tilde{\Lambda}^{n+\frac{1}{2}} D_h \right) \tilde{e}^{n+\frac{1}{2}} &= \frac{2}{\tau_{n+1}} \tilde{e}^n + (S_1 + S_2 \tau_{n+1}) \tilde{e}^n \\
&\quad - (S_1 + S_2 \tau_{n+1}) (\tilde{\Phi}(t_{n+\frac{1}{2}}) - \tilde{\Phi}(t_n)) - \tilde{\Lambda}^{n+\frac{1}{2}} [f(\tilde{\Phi}^n) - f(\tilde{\Phi}(t_{n+\frac{1}{2}}))] \\
&\quad + [\tilde{\Lambda}^{n+\frac{1}{2}} - \Lambda(\tilde{\Phi}(t_{n+\frac{1}{2}}))] [\varepsilon^2 D_h \tilde{\Phi}(t_{n+\frac{1}{2}}) - f(\tilde{\Phi}(t_{n+\frac{1}{2}}))] + \tilde{T}_1^n + \Lambda(\tilde{\Phi}(t_{n+\frac{1}{2}})) \tilde{T}_2^n.
\end{aligned} \tag{3.11}$$

Through a similar analysis as that of (3.5), we can obtain

$$\begin{aligned}
\|\tilde{e}^{n+\frac{1}{2}}\|_{\infty} &\leq \|\tilde{e}^n\|_{\infty} + \left(\frac{S_1}{2} + \frac{S_2 \tau_{n+1}}{2} + M_* \right) \tau_{n+1} \|\tilde{e}^n\|_{\infty} + C_2 \tau_{n+1} \|\tilde{e}^{n+\frac{1}{2}}\|_{\infty} \\
&\quad + \frac{1}{4} (S_1 + S_2 \tau_{n+1} + 2M_*) \|\phi\|_{W^{1,\infty}(0,T;L^{\infty}(\Omega))} \tau_{n+1}^2 \\
&\quad + \frac{1}{8} \|\phi\|_{W^{2,\infty}(0,T;L^{\infty}(\Omega))} \tau_{n+1}^2 + \frac{\varepsilon^2 h^2 M_*}{12} \|\phi\|_{L^{\infty}(0,T;W^{4,\infty}(\Omega))} \tau_{n+1}.
\end{aligned} \tag{3.12}$$

Substituting the estimate (3.10) for $\|\tilde{e}^{n+\frac{1}{2}}\|_{\infty}$ into (3.12), gives

$$\|\tilde{e}^{n+\frac{1}{2}}\|_{\infty} \leq \|\tilde{e}^n\|_{\infty} + C_5 \tau_{n+1} \|\tilde{e}^n\|_{\infty} + C_6 \tau_{n+1}^2 + C_7 \frac{\varepsilon^2 h^2}{12} \|\phi\|_{L^{\infty}(0,T;W^{4,\infty}(\Omega))} \tau_{n+1}, \tag{3.13}$$

where

$$C_5 = \frac{1}{2}(S_1 + S_2\tau + 2M_* + 2(1 + C_3\tau)C_2),$$

$$C_6 = \frac{1}{4}(S_1 + S_2\tau + 2M_*)\|\phi\|_{W^{1,\infty}(0,T;L^\infty(\Omega))} + \frac{1}{8}\|\phi\|_{W^{2,\infty}(0,T;L^\infty(\Omega))} + C_2C_4\tau,$$

and $C_7 = (C_2\tau + 1)M_*$.

Next, we will give an estimate for $\bar{e}^{n+1}$ through the equations (1.1) and (2.9b). The exact solution of the equation (1.1) at $t = t_{n+\frac{1}{2}}$ satisfies

$$\frac{\bar{\Phi}(t_{n+1}) - \bar{\Phi}(t_n)}{\tau_{n+1}} + \Lambda(\bar{\Phi}(t_{n+\frac{1}{2}})) \left[-\varepsilon^2 D_h \frac{\bar{\Phi}(t_{n+1}) + \bar{\Phi}(t_n)}{2} + f(\bar{\Phi}(t_{n+\frac{1}{2}})) \right] + \bar{T}_3^n + \Lambda(\bar{\Phi}(t_{n+\frac{1}{2}}))\bar{T}_4^n = 0,$$

where the local truncation errors $\bar{T}_3^n$ and $\bar{T}_4^n$ are defined by

$$\bar{T}_3^n = \bar{\Phi}(t_{n+\frac{1}{2}}) - \frac{\bar{\Phi}(t_{n+1}) - \bar{\Phi}(t_n)}{\tau_{n+1}}, \quad \bar{T}_4^n = -\varepsilon^2 \Delta \bar{\Phi}(t_{n+\frac{1}{2}}) + \varepsilon^2 D_h \frac{\bar{\Phi}(t_{n+1}) + \bar{\Phi}(t_n)}{2}.$$

It is easy to find that

$$\begin{aligned} \|\bar{T}_3^n\|_\infty &\leq \frac{1}{24}\tau_{n+1}^2 \|\phi\|_{W^{3,\infty}(0,T;L^\infty(\Omega))}; \\ \|\bar{T}_4^n\|_\infty &\leq \varepsilon^2 \left[\frac{\tau_{n+1}^2}{8} \|\phi\|_{W^{3,\infty}(0,T;W^{2,\infty}(\Omega))} + \frac{1}{6}h^2 \|\phi\|_{L^\infty(0,T;W^{4,\infty}(\Omega))} \right] \end{aligned} \quad (3.14)$$

Together with (2.9c), we derive the error equation with respect to $\bar{e}^{n+1}$ as follows:

$$\begin{aligned} &\left(\left(\frac{1}{\tau_{n+1}} + S_2\tau_{n+1} + \frac{S_1}{2} \right) I - \frac{\varepsilon^2}{2} \bar{\Lambda}^{n+\frac{1}{2}} D_h \right) \bar{e}^{n+1} \\ &= Q^{n+1} \bar{e}^n + S_1 \bar{e}^{n+\frac{1}{2}} - S_2 \tau_{n+1} (\bar{\Phi}(t_{n+1}) - \bar{\Phi}(t_n)) \\ &\quad + S_1 \left(\bar{\Phi}(t_{n+\frac{1}{2}}) - \frac{\bar{\Phi}(t_{n+1}) + \bar{\Phi}(t_n)}{2} \right) - \bar{\Lambda}^{n+\frac{1}{2}} [f(\bar{\Phi}^{n+\frac{1}{2}}) - f(\bar{\Phi}(t_{n+\frac{1}{2}}))] \\ &\quad - [\bar{\Lambda}^{n+\frac{1}{2}} - \Lambda(\bar{\Phi}(t_{n+\frac{1}{2}}))] [-\varepsilon^2 D_h \bar{\Phi}(t_{n+\frac{1}{2}}) + f(\bar{\Phi}(t_{n+\frac{1}{2}}))] + \bar{T}_3^n + \Lambda(\bar{\Phi}(t_{n+\frac{1}{2}}))\bar{T}_4^n \\ &:= Q^{n+1} \bar{e}^n + S_1 \bar{e}^{n+\frac{1}{2}} + R^n, \end{aligned} \quad (3.15)$$

where R^n represents the remaining terms in the right-hand side except for the first two terms and the matrix Q^{n+1} is given by

$$Q^{n+1} = \left(\frac{1}{\tau_{n+1}} - \frac{S_1}{2} + S_2\tau_{n+1} \right) I + \frac{\varepsilon^2}{2} \bar{\Lambda}^{n+\frac{1}{2}} D_h.$$

We deduce from (2.12) that

$$Q^{n+1} \geq \left(2\sqrt{S_2} - \frac{S_1}{2} \right) I + \frac{\varepsilon^2}{2} \bar{\Lambda}^{n+\frac{1}{2}} D_h \geq \frac{M_*\varepsilon^2}{h^2} I + \frac{\varepsilon^2}{2} \bar{\Lambda}^{n+\frac{1}{2}} D_h > 0,$$

which means that all elements of Q are positive. Thus, it gives that

$$\begin{aligned} \|Q^{n+1}\|_\infty &= \max_{1 \leq i \leq M^2} \sum_{j=1}^{M^2} \left| \left(\left(\frac{1}{\tau_{n+1}} + S_2\tau_{n+1} - \frac{S_1}{2} \right) I + \frac{\varepsilon^2}{2} \bar{\Lambda}^{n+\frac{1}{2}} D_h \right)_{i,j} \right| \\ &\leq \frac{1}{\tau_{n+1}} + S_2\tau_{n+1} - \frac{S_1}{2}, \end{aligned}$$

where we have used the fact $\sum_{j=1}^{M^2} \left(\bar{\Lambda}^{n+\frac{1}{2}} D_h \right)_{i,j} = 0$ for any $1 \leq i \leq M^2$. We then obtain

$$\|Q^{n+1} \bar{e}^n\|_\infty \leq \left(\frac{1}{\tau_{n+1}} + S_2\tau_{n+1} - \frac{S_1}{2} \right) \|\bar{e}^n\|_\infty \leq \left(\frac{1}{\tau_{n+1}} + S_2\tau_{n+1} + \frac{S_1}{2} \right) \|\bar{e}^n\|_\infty. \quad (3.16)$$

Similarly to (3.9), we use the definition of $\bar{T}_4^n$ to derive that

$$\begin{aligned}
& \| -\varepsilon^2 D_h \bar{\Phi}(t_{n+\frac{1}{2}}) + F'(\bar{\Phi}(t_{n+\frac{1}{2}})) \|_\infty \\
& \leq \varepsilon^2 \|\phi\|_{L^\infty(0,T;W^{2,\infty}(\Omega))} + \frac{2}{3\sqrt{3}} + \varepsilon^2 \left(\frac{\tau^2}{8} \|\phi\|_{W^{3,\infty}(0,T;W^{2,\infty}(\Omega))} + \frac{h^2}{6} \|\phi\|_{L^\infty(0,T;W^{4,\infty}(\Omega))}^2 \right) \\
& := C_8
\end{aligned} \tag{3.17}$$

Combining (3.15) with (3.16) and (3.17), we apply Lemma 2.2 and the triangle inequality to derive

$$\begin{aligned}
\|\bar{e}^{n+1}\|_\infty & \leq \left\| \left(\left(\frac{1}{\tau_{n+1}} + S_2 \tau_{n+1} + \frac{S_1}{2} \right) I - \frac{\varepsilon^2}{2} \bar{\Lambda}^{n+\frac{1}{2}} D_h \right) \bar{e}^{n+1} \right\|_\infty \\
& = \tau_{n+1} \|\bar{Q}^{n+1} \bar{e}^n + S_1 \bar{e}^{n+\frac{1}{2}} + R^n\|_\infty \\
& \leq \|\bar{e}^n\|_\infty + \left(\frac{S_1}{2} + S_2 \tau_{n+1} \right) \tau_{n+1} \|\bar{e}^n\|_\infty + S_1 \tau_{n+1} \|\bar{e}^{n+\frac{1}{2}}\|_\infty \\
& \quad + \frac{1}{8} S_1 \tau_{n+1}^3 \|\phi\|_{W^{2,\infty}(0,T;L^\infty(\Omega))} + S_2 \tau_{n+1}^3 \|\phi\|_{W^{1,\infty}(0,T;L^\infty(\Omega))} + 2 \tau_{n+1} M_* \|\bar{e}^{n+\frac{1}{2}}\|_\infty \\
& \quad + C_8 \max_{\rho \in [-1,1]} |M'(\rho)| \tau_{n+1} \|\bar{e}^{n+\frac{1}{2}}\|_\infty + \frac{1}{24} \|\phi\|_{W^{3,\infty}(0,T;L^\infty(\Omega))} \tau_{n+1}^3 \\
& \quad + \varepsilon^2 \tau_{n+1} M_* \left[\frac{\tau_{n+1}^2}{8} \|\phi\|_{W^{3,\infty}(0,T;W^{2,\infty}(\Omega))} + \frac{1}{6} h^2 \|\phi\|_{L^\infty(0,T;W^{4,\infty}(\Omega))} \right] \\
& \leq \|\bar{e}^n\|_\infty + \left(\frac{S_1}{2} + S_2 \tau_{n+1} \right) \tau_{n+1} \|\bar{e}^n\|_\infty + (S_1 + 2M_*) \tau_{n+1} \|\bar{e}^{n+\frac{1}{2}}\|_\infty \\
& \quad + C_8 M_* \tau_{n+1} \|\bar{e}^{n+\frac{1}{2}}\|_\infty + C_9 \tau_{n+1}^3 + \tau_{n+1} \frac{\varepsilon^2 h^2 M_*}{6} \|\phi\|_{L^\infty(0,T;W^{4,\infty}(\Omega))}
\end{aligned} \tag{3.18}$$

where

$$\begin{aligned}
C_9 & = \frac{1}{8} S_1 \|\phi\|_{W^{2,\infty}(0,T;L^\infty(\Omega))} + S_2 \|\phi\|_{W^{1,\infty}(0,T;L^\infty(\Omega))} \\
& \quad + \frac{1}{24} \|\phi\|_{W^{3,\infty}(0,T;L^\infty(\Omega))} + \frac{M_*}{8} \|\phi\|_{W^{3,\infty}(0,T;W^{2,\infty}(\Omega))}.
\end{aligned}$$

Substituting the error estimates for $\|\bar{e}^{n+\frac{1}{2}}\|_\infty$ from (3.10) and $\|\bar{e}^{n+\frac{1}{2}}\|_\infty$ from (3.13) into (3.18), yields

$$\|\bar{e}^{n+1}\|_\infty \leq \|\bar{e}^n\|_\infty + \tau_{n+1} \left(C^* \|\bar{e}^n\|_\infty + C_{10} \tau_{n+1}^2 + C_{11} h^2 \right) \tag{3.19}$$

where

$$\begin{aligned}
C^* & = \frac{S_1}{2} + S_2 \tau_{n+1} + (1 + C_5 \tau_{n+1})(S_1 + 2M_*) + (1 + C_3 \tau_{n+1}) C_8 M_* \tau, \\
C_{10} & = (S_1 + 2M_*) C_6 + M_* C_4 C_8 + C_9, \\
C_{11} & = \left((S_1 + 2M_*) C_7 \tau_{n+1} + M_* C_8 \tau_{n+1} + 2 \right) \frac{\varepsilon^2}{12} \|\phi\|_{L^\infty(0,T;W^{4,\infty}(\Omega))}.
\end{aligned}$$

Finally we obtain

$$\|\bar{e}^{n+1}\|_\infty \leq \|\bar{e}^n\|_\infty + \tau_{n+1} \left(C^* \|\bar{e}^n\|_\infty + C(\tau_{n+1}^2 + h^2) \right) \tag{3.20}$$

where $C = \max\{C_{10}, C_{11}\}$. Summing up (3.20) from 0 to n , and together with the discrete the Gronwall's Lemma 3.1, the desired estimate (3.1) can be obtained, which completes the proof. $\square$

Remark 3.1. For the logarithmic potential case, the potential function $F(\cdot)$ satisfies the Assumption 2 in [10], i.e., there exists a positive constant β such that $F'(\beta) \leq F'(-\beta)$. Consequently, the results obtained in Lemma 2.3 hold for any $\rho \in [-\beta, \beta]$. Thus the proposed scheme can be directly applied to the case of the logarithmic potential function, unconditionally preserving the discrete maximum bound principle (MBP) and the original energy dissipation. Thanks to the MBP preservation, the corresponding error estimates also can be derived in a manner similar to Theorem 2.3. When the proposed scheme is extended to the Cahn-Hilliard equation, by assuming the nonlinear term $f = F'$ satisfies the Lipschitz condition, one can obtain analogous analysis results as presented in [13], including the preservation of the original energy dissipation law and the error estimates.

4. Numerical experiments

In this section, we present a series of numerical experiments to validate the theoretical results for the proposed sCN scheme (2.8) in terms of accuracy and unconditional preservation of the discrete MBP and original energy dissipation law. Throughout the numerical

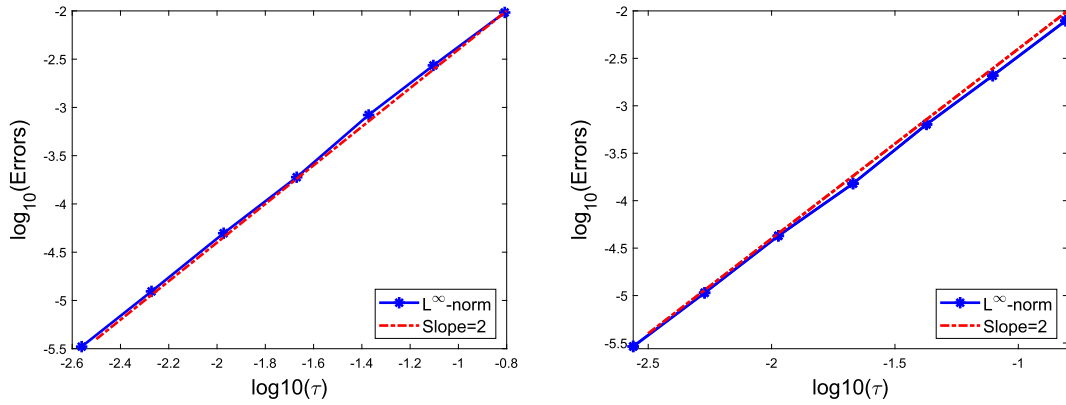


Fig. 1. The error behavior of the proposed sCN scheme (2.8) based on uniform time steps. Left: $M(\phi) \equiv 1$; Right: $M(\phi) = 1 - \phi^2$.

Table 1

Numerical solution errors and convergence behavior of the sCN scheme (2.8) based on nonuniform time steps.

Time steps			$M(\phi) = 1$		$M(\phi) = 1 - \phi^2$	
N	τ	$\max\{\gamma_n\}$	e_∞^T	Order	e_∞^T	Order
10	1.356e-01	3.441	9.630e-03	-	7.412e-03	-
20	7.387e-02	2.403	2.723e-03	2.36	2.089e-03	2.35
40	4.474e-02	5.608	8.351e-04	1.95	6.429e-04	1.93
80	2.087e-02	5.437	1.892e-04	1.90	1.473e-04	1.87
160	1.034e-02	3.750	4.972e-05	2.13	3.968e-05	2.06
320	5.396e-03	4.608	1.242e-05	1.95	1.043e-05	1.81

tests, the models are all subject to homogeneous Neumann boundary conditions. These experiments also aim to demonstrate the effectiveness and reliability of the proposed scheme in practical applications.

4.1. Convergence tests

For the Allen-Cahn equation with general mobility, we consider two types of cases: one with constant mobility $M(\phi) = 1$ and another with nonlinear mobility $M(\phi) = 1 - \phi^2$. We aim to verify the accuracy of these two cases, where the parameter ε of the Allen-Cahn equation is set to 0.01 and the initial condition is given by

$$\phi_0(x, y) = 0.1(\cos 3x \cos 2y + \cos 5x \cos 5y), \quad (x, y) \in \Omega := (0, 1)^2.$$

The terminal time is set to $T = 1$, and the stabilization parameters S_1 and S_2 are both set to 2. A sufficiently small spatial mesh size of $h = 1/1024$ is used to ensure that the spatial discretization error is negligible compared to the temporal approximation error. Since there is no analytical solution available for this numerical experiment, the L^∞ -norm of the numerical solution errors are evaluated as follows:

$$e_\infty^T(N) = \|\Phi^N - \Phi^{2N}\|_\infty,$$

where Φ^N and Φ^{2N} denote the terminal numerical solutions at T obtained by dividing the time domain $[0, T]$ into N and $2N$ subintervals, respectively.

First, we evaluate the temporal convergence rate of the sCN scheme (2.8) using uniform temporal meshes with time step sizes τ ranging from $1/10$ to $1/320$, which corresponds to N varying from 10 to 320. We plot the L^∞ errors as functions of the time step sizes in log-log scale in Fig. 1, which shows the expected second-order temporal accuracy of the proposed sCN scheme (2.8) for both mobility cases. Next, we numerically explore the error behavior of the sCN scheme (2.8) on nonuniform temporal meshes, generated by applying a 40% perturbation to the uniform ones $\{t_n = n/N\}_{n=0}^N$. Let $\{\gamma_n = \Delta t_n / \Delta t_{n-1}\}_{n=2}^N$ denote the adjacent time-step ratios. Furthermore, the convergence order of the numerical scheme is evaluated by:

$$\text{Order} \approx \frac{\log \frac{\text{error}(N)}{\text{error}(2N)}}{\log \frac{\tau(N)}{\tau(2N)}}.$$

Once again, the error behavior observed in Table 1 demonstrates the desired second-order accuracy in time for all tested nonuniform time steps.

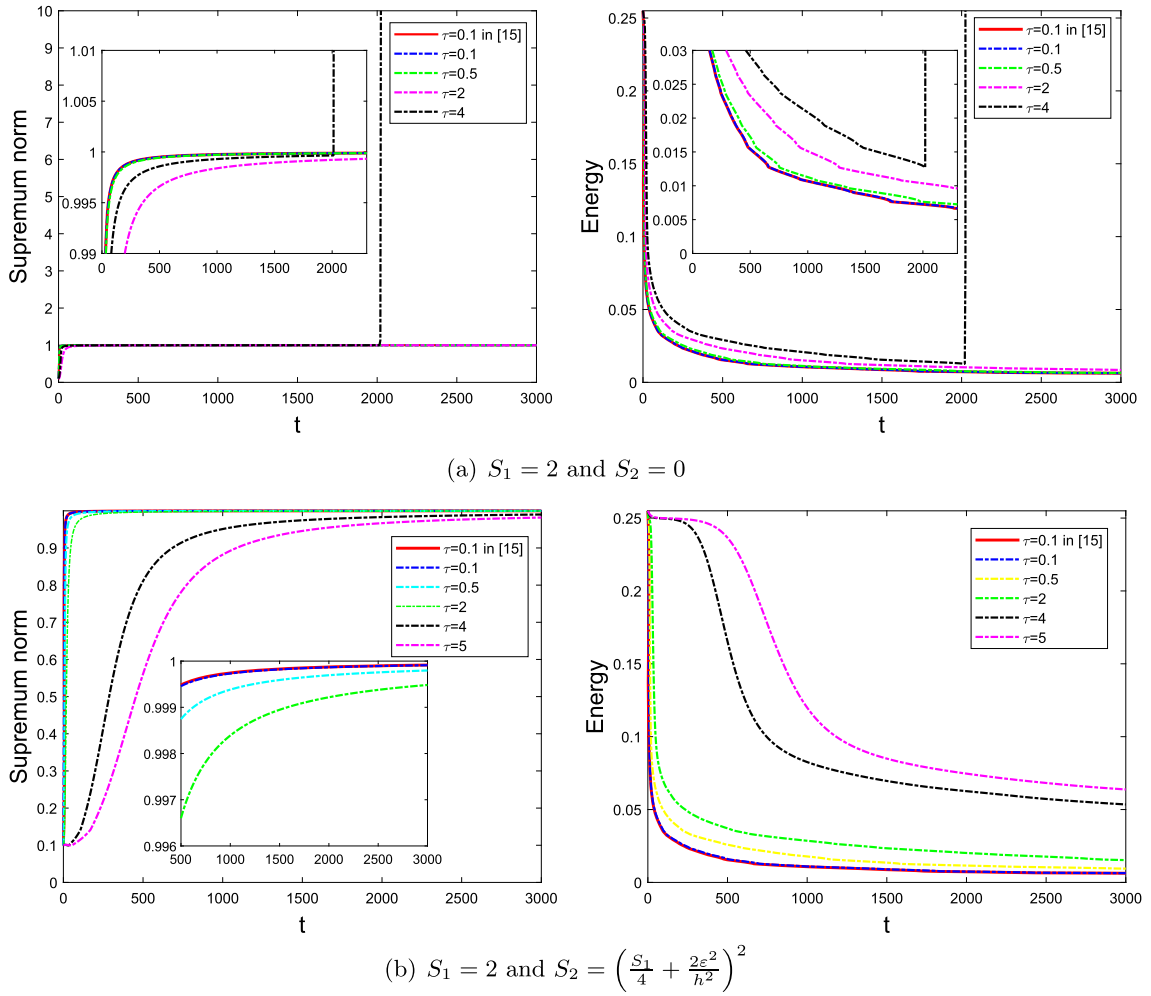


Fig. 2. The evolutions in time of the supremum norm (left) and the energy (right) of the simulated solution produced by the sCN scheme (2.8) with several uniform time steps for the grain coarsening problem with the degenerate mobility.

4.2. MBP preservation and original energy dissipation

To validate the unconditional preservation of the discrete MBP and energy dissipation law for the proposed sCN scheme (2.8), we examine two well-known benchmark problems described by the Allen-Cahn equations: the grain coarsening dynamic process and the shrinking bubble problem [5].

4.2.1. The grain coarsening dynamics in 2D

We first investigate the sCN scheme (2.8) for the Allen-Cahn equation model (1.1) in the computational domain $\Omega = (0, 1)^2$ with $\varepsilon = 1/256 \approx 3.91 \times 10^{-3}$. We consider the model with the degenerate mobility $M(\phi) = 1 - \phi^2$ and the random initial data within the range $[-0.1, 0.1]$, the same as the example studied in [16]. For such a nonlinear mobility function, it is essential to ensure that the numerical solution ϕ^n remains within the interval $[-1, 1]$. Otherwise, the proposed numerical scheme may become ill-posed, and the numerical solution may blow up during the time simulation. Long-time simulations up to $T = 3000$ are performed using the proposed sCN scheme (2.8) with a spatial grid size $h = 1/256$ and several different large time step sizes as shown in Fig. 2. The evolutions of the supremum norm and the discrete original energy of the simulated solutions are plotted for two sets of stabilizing parameters: $S_1 = 2$ satisfying the condition (2.10) with $S_2 = 0$, and $S_1 = 2$ with $S_2 = \left(\frac{S_1}{4} + \frac{2\varepsilon^2}{h^2}\right)^2$ satisfying (2.19). As shown in Fig. 2(a), the proposed sCN scheme with the stabilizing parameters $S_1 = 2$ and $S_2 = 0$ preserves the discrete MBP and energy dissipation law very well for the tested time step sizes $\tau = 0.1, 0.5, 2$, while the numerical scheme developed in [16] with $\tau = 2$ blows up. When $\tau = 4$, our sCN scheme also blows up at the time around $t = 2000$. In the case where the stabilizing parameters $S_1 = 2$ and $S_2 = \left(\frac{S_1}{4} + \frac{2\varepsilon^2}{h^2}\right)^2$, our sCN scheme preserve the discrete MBP and original dissipation law for all tested time step sizes, even when $\tau = 5$. These numerical results are consistent well with the theoretical results presented in Theorem 2.1.

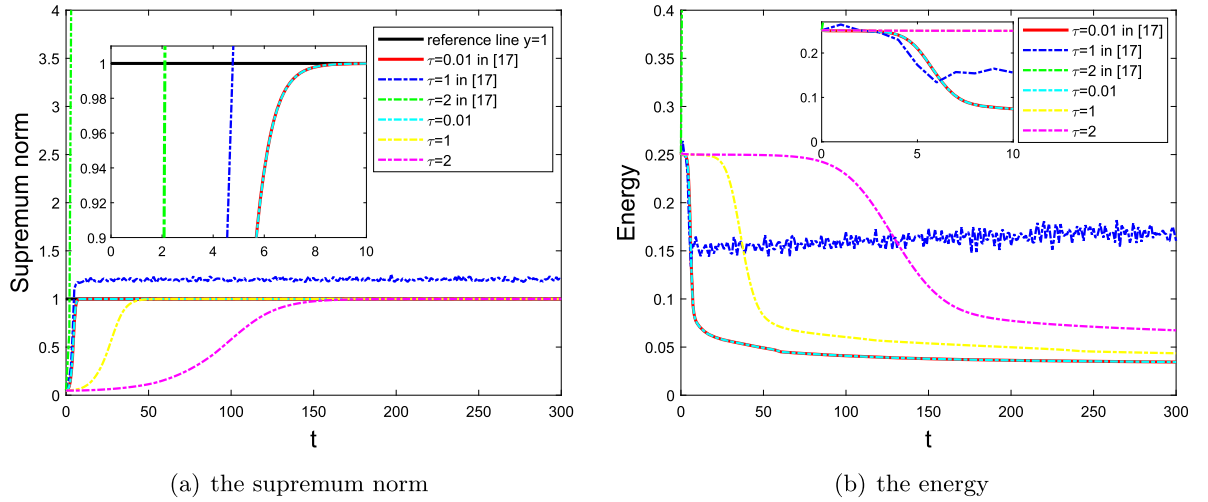


Fig. 3. The evolutions in time of the supremum norm (left) and the original energy (right) produced by the sCN scheme (2.8) and the CN scheme in [17].

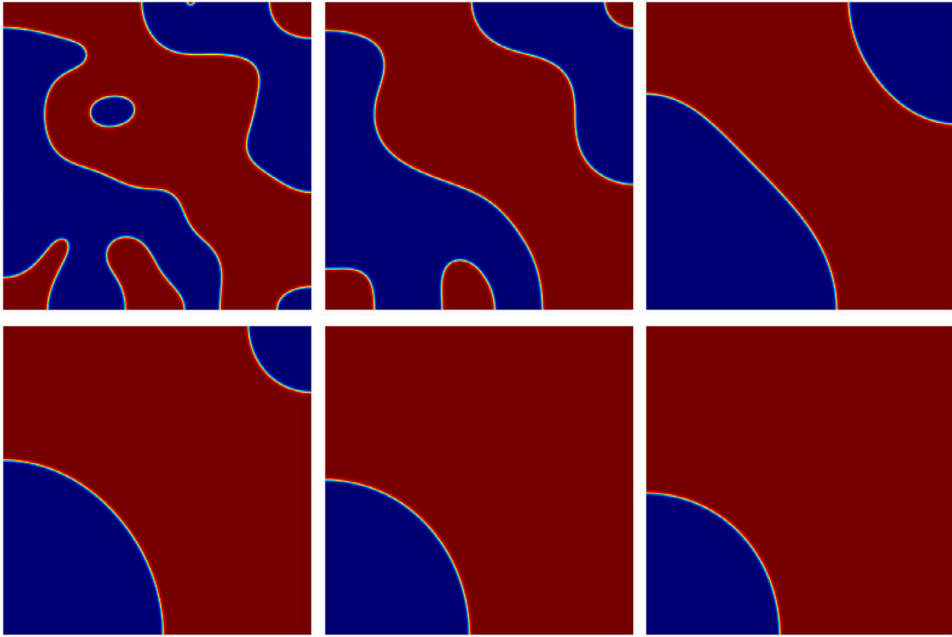


Fig. 4. Snapshots of the simulated phase structures around the times $t = 500, 1000, 5000, 10000, 13000$, and 15000 produced by the sCN scheme (2.8) with the time adaptive strategy (4.1) for the grain coarsening problem with the degenerate mobility in 2D.

Next, we investigate the sCN scheme (2.8) for the Allen-Cahn equation model (1.1) with a constant mobility $M(\phi) = 1$ in the computational domain $\Omega = (0, 1)^2$ with $\varepsilon = 0.01$, and the random initial data within the range $[-0.05, 0.05]$. The simulations up to $T = 300$ are performed with a spatial grid size $h = 1/100$. Under identical settings, we compare our scheme with the Crank-Nicolson/Adams-Bashforth finite difference scheme reported in [17]. As shown in Fig. 3, the evolutions of the supremum norm and original energy computed by the sCN scheme (2.8) and the CN scheme in [17] are consistent when a small time step $\tau = 0.01$ is used. However, the numerical scheme in [17] fails to preserve the discrete MBP and the original energy dissipation for large time step sizes $\tau = 1$ and $\tau = 2$, even exhibiting blow-up phenomena when $\tau = 2$. In contrast, our scheme demonstrates superior stability while preserving both the discrete MBP and the original energy dissipation for the large time step sizes $\tau = 1$ and $\tau = 2$.

One of the primary advantages of unconditionally stable schemes is their ease of integration with adaptive time strategies. This is especially beneficial for long-term simulations of coarsening dynamic processes, as phase transitions typically go through several different stages over an extended period: rapid changes at first, followed by a gradual slowdown until a steady state is reached. Therefore, we will utilize the following time adaptation strategy based on energy changes [32], combined with the proposed sCN scheme (2.8), to effectively simulate the coarsening dynamic process:

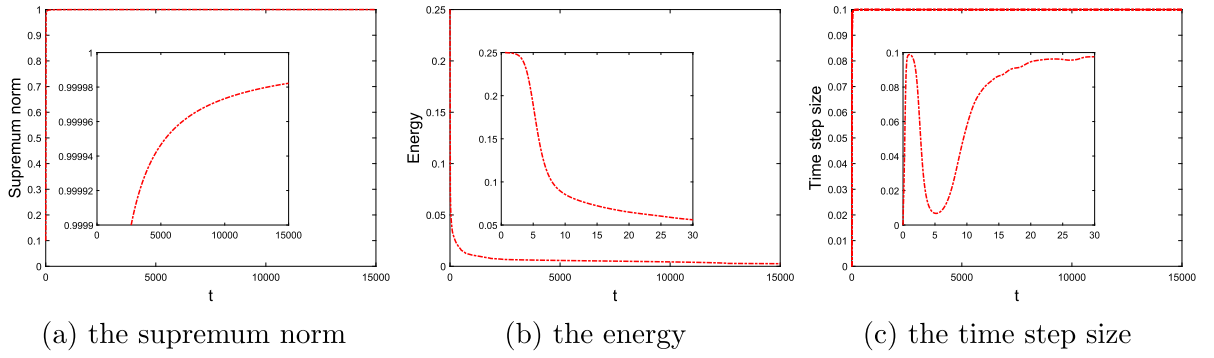


Fig. 5. The evolutions in time of the supremum norm, the energy, and the time step sizes produced by the sCN scheme (2.8) with the time adaptive strategy (4.1) for the grain coarsening problem with the degenerate mobility.

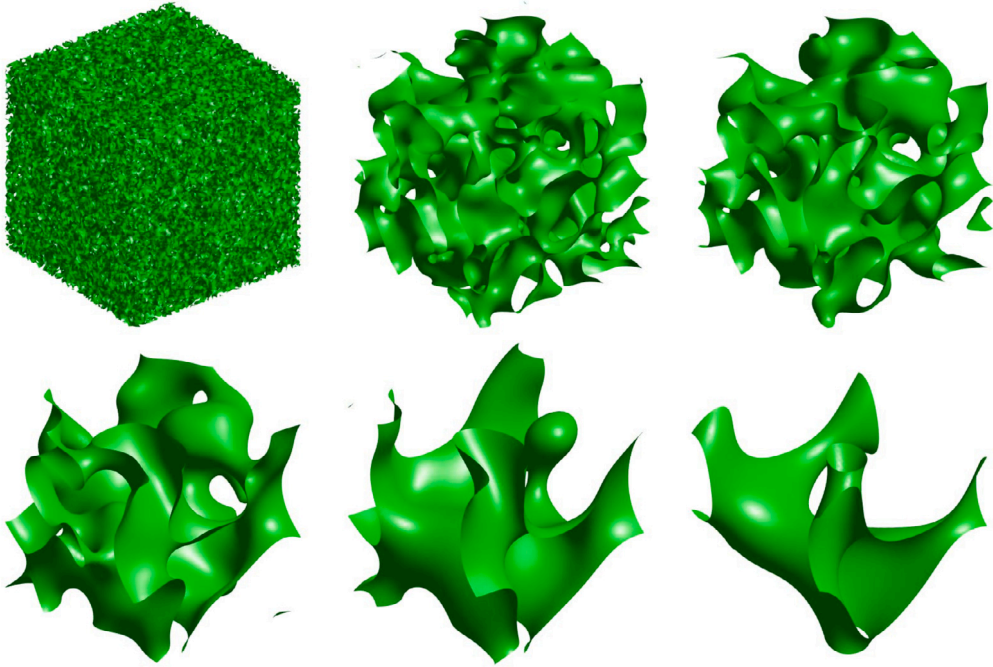


Fig. 6. Snapshots of the simulated phase structures at the times $t = 0, 10, 20, 50, 100$, and 200 showing coarsening dynamics.

$$\tau_{n+1} = \max \left(\tau_{min}, \frac{\tau_{max}}{\sqrt{1 + \alpha |E'(t)|^2}} \right), \quad (4.1)$$

where τ_{min} and τ_{max} are the predetermined minimum and maximum time step sizes, respectively. Here, α is a positive constant parameter, set to $\alpha = 10^5$ in this test. In the simulation, we set $\tau_{min} = 10^{-5}$, $\tau_{max} = 0.1$, $S_1 = 2$ and $S_2 = \left(\frac{S_1}{4} + \frac{2\epsilon^2}{h^2} \right)^2$ to ensure that both conditions (2.10) and (2.19) are satisfied. We display snapshots of the simulated phase structures around the times $t = 500, 1000, 5000, 10000, 13000, 15000$ in Fig. 4. It is observed in Fig. 5(a) and Fig. 5(b) that the sCN scheme (2.8) always preserves the MBP and the original energy dissipation law throughout the simulation up to the terminal time $T = 15000$. Moreover, we also plot the evolution of the adaptive time step sizes as a function of time in Fig. 5(c), where the adaptive time stepping strategy automatically selects the time step sizes according to the changing rate of the free energy, which demonstrates the efficiency of our proposed sCN scheme in combination with the time adaptive strategy (4.1).

4.2.2. The gain coarsening dynamics in 3D

We consider the Allen-Cahn equation (1.1) with mobility $M(\phi) = 1 - \phi^2$ and $\epsilon = \frac{1}{64}$, subject to the periodic boundary condition. The computational domain is chosen as $\Omega = (-0.5, 0.5)^3$, with an initial data given by random values in $(-0.9, 0.9)$.

The simulation is performed by the sCN scheme (2.8) with a fixed time step size $\tau = 0.1$ and a uniform grid spacing size $h = \frac{1}{64}$ up to final time $T = 200$. The evolution of the numerical solution is shown in Fig. 6, demonstrating the coarsening dynamics. As shown, the initially disordered phase field gradually evolves, exhibiting significant morphological differences between the initial state and

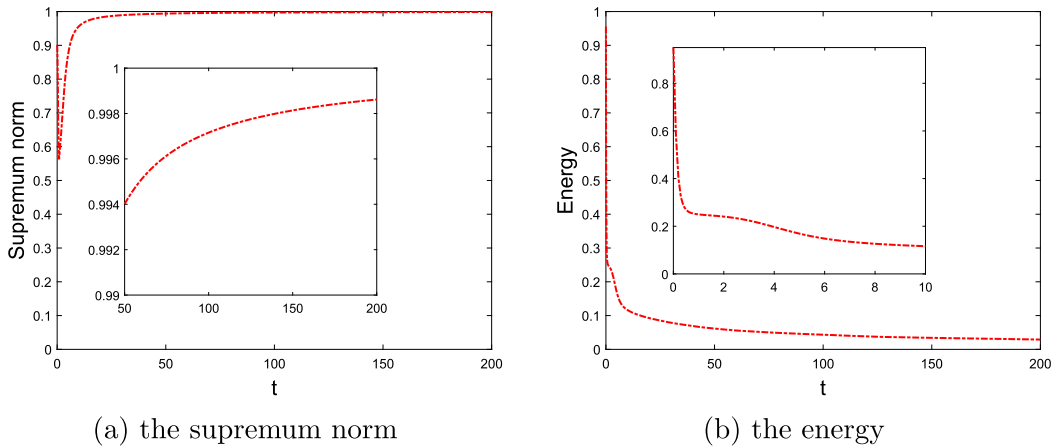


Fig. 7. The evolutions in time of the supremum norm and the energy produced by the sCN scheme (2.8) for the grain coarsening problem with the degenerate mobility in 3D.

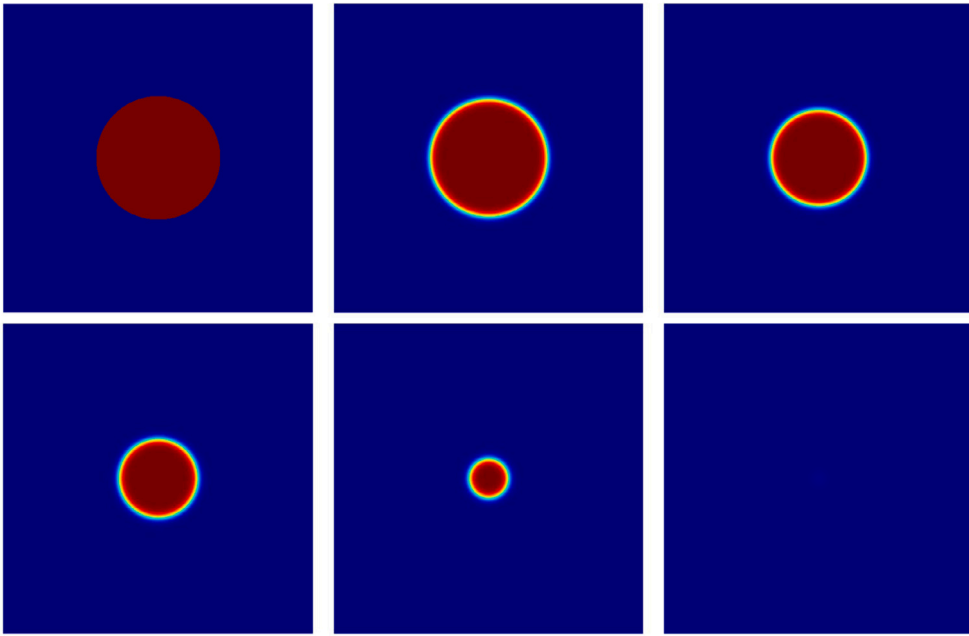


Fig. 8. Snapshots of the simulated phase structures at the times $t = 0, 20, 80, 120, 180$, and 200 produced by the CN scheme (2.8) for the shrinking bubble problem.

the configuration at $T = 200$. Moreover, it shows in Fig. 7 that our sCN scheme preserves both the discrete MBP and original energy dissipation in the tested 3D case.

4.2.3. The shrinking bubble problem

We next consider the shrinking bubble problem governed by the Allen-Cahn equation (1.1) in the rectangular domain $\Omega = (-0.5, 0.5)^2$, with $M(\phi) \equiv 1$ and $\varepsilon = 0.01$. The initial bubble is a sphere with a radius of $R_0 = 0.2$, centered in the computational domain, and is defined as follows: As described in [6,14,24], the model describes the evolution of a bubble shrinking and eventually disappearing over time. If ε is sufficiently small, the radius of this circle at time t is approximated as follows

$$R(t) = \sqrt{R_0^2 - 2\varepsilon^2 t}. \quad (4.2)$$

The simulation is carried out using the sCN scheme (2.8) with $h = 1/512$ and the adaptive time strategy (4.1), with $\tau_{min} = 10^{-5}$, $\tau_{max} = 0.01$, and $\alpha = 10^5$. Snapshots of the simulated bubble at about $t = 0, 20, 80, 120, 180, 200$ are shown in Fig. 8. The plot of the bubble's radius over time in Fig. 9 (a) aligns well with the prediction (4.2), which indicates that the bubble disappears at $t = 200$. Fig. 9 (b)-(d) present the evolution of the supremum norm, energy, and time step sizes, respectively. It shows the proposed CN scheme (2.8) preserve the discrete MBP and original energy dissipation law throughout the simulation.

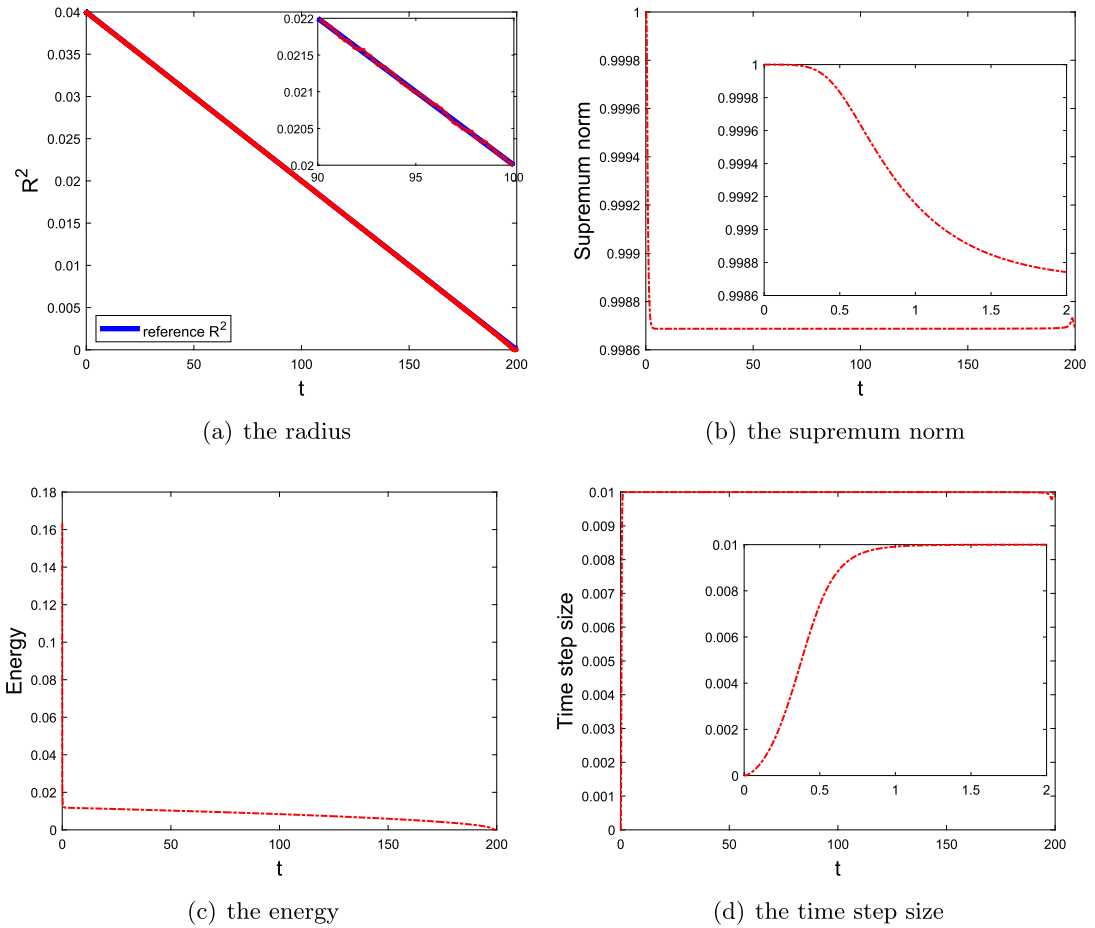


Fig. 9. The evolutions in time of the radius, the supremumnorm, the energy, and the time step sizes produced by the sCN scheme ((2.8) with the time adaptive strategy (4.1) for the shrinking bubble problem.

5. Concluding remarks

In this paper we have developed a linear Crank-Nicolson scheme for the Allen-Cahn equation with general mobility, in which two types of stabilizing terms are incorporated. We have rigorously established the unconditional preservation of both the discrete MBP and the discrete original energy dissipation law. Additionally, we have derived fully-discrete L^∞ error estimates for the proposed scheme. Numerical experiments are conducted to validate the theoretical results and to demonstrate the scheme's performance, particularly when combined with an adaptive time-stepping strategy. Looking ahead, we aim to further analyze the energy dissipation properties of the widely used BDF2 scheme with similar stabilizing terms for general gradient flows, with special focus on the Allen-Cahn and Cahn-Hilliard equations.

Data availability

Data will be made available on request.

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